Outcome-Flip Sensitivity of Pairwise Leaderboards: An Empirical Audit Across Evaluation Designs

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Abstract

Pairwise leaderboards turn comparison outcomes into an ordered list, but the stability of that list depends on both the observed evidence and the fitted model. We audit retrospective outcome-flip sensitivity in public Chatbot Arena preferences, MT-Bench human and GPT-4 judgments, and ten NBA regular seasons. We reverse distinct decisive outcomes while preserving comparison schedules and ties, then refit a ridge Bradley–Terry model. Influence-guided searches find reversals of the original leading pair after 2 of 57,477 Arena outcomes, 46 of 3,355 MT-Bench human judgments, 61 of 2,400 GPT-4 judgments, and 1–8 games per NBA season. These are successful-edit upper bounds, not minimum budgets. Arena and every NBA leading-pair bootstrap interval already include zero; the two MT-Bench intervals exclude it. On identical MT-Bench comparison cells, a paired question bootstrap gives a mean human-minus-GPT-4 found edit fraction of -1.20 percentage points, with a descriptive percentile range of $[-2.24, -0.11]$ points for a fixed reference pair. Sensitivity also changes with tie handling and regularization. The results support reporting edit witnesses alongside sampling uncertainty and modeling conventions; they do not establish that an evaluation design is inherently more vulnerable or that retrospective flips are operational attacks.

1 Introduction

A pairwise leaderboard can display a precise ordering even when the evidence separating neighboring entries is weak. This matters when a first-place label informs model selection, publicity, or scientific comparison. A small change to an outcome may have a negligible effect on fitted scores yet reverse a nearly tied pair. Conversely, a leaderboard with a modest sample can retain its leading contrast under many changes if that contrast is well separated. Counting edits alone therefore does not explain what a rank reversal means.

We study the following operational question: how do fitted rankings change when a small number of observed decisive outcomes are reversed, with the comparison schedule held fixed? Such an intervention can represent a sensitivity diagnostic for miscoded outcomes or deliberately chosen changes. It does not assume that anyone can edit an actual platform's history. We compare uniform random reversals, a simple targeted baseline, and an influence-guided search, measuring all reported effects with full model refits.

Our evidence covers crowd votes, human judgments on a fixed benchmark, model judgments on that benchmark, and sports results. These are deliberately different settings. We do not interpret the differences as effects of evaluation design: sample size, competitors, gaps, ties and schedules vary together. A matched MT-Bench comparison narrows the question by fixing the observed question–pair–turn cells and comparing label sources on the same support.

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This paper contributes an empirical audit with three linked measurements: explicit successful edits to a selected leading contrast, random-perturbation and whole-ranking responses at matched fractional budgets, and cluster-bootstrap uncertainty before any edits. We also examine regularization, tie conventions, entrant coverage, human annotation subsets, and home advantage. The contribution is the jointly qualified evidence across these snapshots, rather than a new estimator or an optimal attack algorithm. In particular, "two edits" describes one fitted Arena snapshot under one convention; it is not a universal robustness threshold.

2 Related work

Bradley–Terry models aggregate pairwise outcomes into latent strengths [Chiang et al., 2024]. Chatbot Arena operationalizes human preference comparisons for language-model evaluation [Chiang et al., 2024], while MT-Bench provides human and model judgments that permit comparisons between label sources [Zheng et al., 2023]. We analyze the supplied public snapshots rather than reconstruct the platforms' current production rankings. In particular, our unweighted empirical-schedule ridge fit differs from Arena's reported inverse-sampling-weighted estimation for a uniform pair distribution, and our selected-gap bootstrap does not reproduce its simultaneous ranking-uncertainty procedure.

Influence functions approximate how a fitted quantity responds to small changes in observations [Koh and Liang, 2017]. Finite-sample robustness work asks whether changes to a small portion of a dataset can alter a substantive conclusion [Broderick et al., 2020]. Our use is narrower: a first-order score ranks candidate outcome reversals, and exact refits establish the resulting selected-pair changes. The candidate ordering is a computational heuristic, not a global robustness certificate. We preserve rows and participants, so results for deletion, reweighting, or vote injection are not interchangeable with this intervention. Huang et al. study influence-guided deletion sensitivity in Bradley–Terry leaderboards and validate candidate deletions by refitting [Huang et al., 2025]. We therefore do not claim the first demonstration of leaderboard sensitivity or introduce influence-guided ranking audits. Our outcome-flip intervention retains ties and comparison schedules, uses a specified ridge fit, adds sports seasons, and places successful witnesses beside random perturbations and baseline uncertainty. Min et al. study new-vote injection under platform information and access constraints [Min et al., 2025]; changing existing labels in a frozen dataset does not evaluate that operational threat model. Evaluation-policy questions, including selective disclosure and unequal sampling, are another distinct concern [Singh et al., 2025]; our fixed-schedule experiments do not test those mechanisms. Published deletion or injection thresholds are not treated as newly executed numerical baselines.

3 Question, estimands, and method

Data and fitting convention. An observation contains two participants a_i, b_i and a canonical outcome $y_i \in \{0, \frac{1}{2}, 1\}$. The first participant wins when $y_i = 1$; a tie contributes half a win. For $x_i = e_{a_i} - e_{b_i}$ and $p_i = \sigma(x_i^\top \theta)$, we minimize

$$L(\theta) = \sum_{i=1}^{N} \left[\log(1 + e^{x_i^\top \theta}) - y_i x_i^\top \theta \right] + \frac{\lambda}{2} \|\theta\|_2^2. \tag{1}$$

We use $\lambda = 1$ and centered logit scores in the primary study. The loss is summed, not averaged, so this convention must be specified when comparing sample sizes. Ties are fractional outcomes, not a fitted tie-probability model. Positive ridge yields a unique fit and controls separated or weakly supported competitors. Newton updates with line search are checked for convergence; every reported primary comparison graph is connected.

Intervention and budget. Choose distinct decisive rows S and replace y_i with $1 - y_i$ for $i \in S$. Participants, row count, schedule, and all ties remain unchanged. We report the absolute count $k = |S|$ and k/N, where N includes ties. Using all rows makes the denominator explicit across datasets with different tie rates; it does not make their information content equivalent. We also provide decisive-row counts. Curve budgets are one edit and $\max(1, \lfloor qN \rfloor)$ for $q \in \{0.001, 0.0025, 0.005, 0.01, 0.02, 0.05\}$. Candidate exhaustion uses at most the stated budget and records the actual count.

Conditional and global endpoints. Select the initial leader ℓ and runner-up r once from the original fit. The primary endpoint is reversal of $g = (e_\ell - e_r)^\top \hat{\theta}$: after refitting, $\theta_r > \theta_\ell$ (numerical tolerance 10^{-10}). Reversal necessarily displaces the original leader, but r need not become the new overall winner; a third participant may do so. Conversely, a winner change can occur without this pair reversing. We separately record whether any participant displaces the initial leader, its new rank, top-five overlap, and normalized Kendall disagreement $D = (1 - \tau)/2$. The latter summarizes all pairwise ordering changes and is descriptive across different participant counts.

Candidate searches and baselines. Let $c = e_\ell - e_r$ and

$$H = \sum_i p_i(1 - p_i)x_i x_i^\top + \lambda I. \tag{2}$$

The first-order change in g from a reversal is

$$\delta_i = (1 - 2y_i)c^\top H^{-1} x_i. \tag{3}$$

We sort decisive rows with $\delta_i < 0$ by decreasing $-\delta_i$. Prefixes are evaluated by exact refits, with a search cap of $\lfloor 0.05N \rfloor$. Binary search locates a successful prefix when the cap succeeds; the delivered audit records both its margin and that of the adjoining smaller prefix. We call its count a *found budget*. No monotonicity proof or search over all subsets is supplied. A successful witness establishes an upper bound on the minimum number needed for this endpoint; failure of a searched prefix establishes no lower bound.

Uniform random reversal samples decisive rows without replacement, with 500 independent permutations per primary ranking and nested budgets. The endpoint baseline randomly orders decisive rows involving the leader or runner-up whose direct sign weakens the original contrast, with 50 permutations. This tests whether simple targeting already suffices; influence-guided performance is not assumed superior for every dataset or metric. Neither baseline optimizes whole-ranking disagreement. Representative single-row refits check the influence approximation, while full-budget refits prevent linear extrapolation from becoming reported evidence.

Two kinds of uncertainty. Random-edit variation concerns a perturbation strategy conditional on the observed data. We report exact binomial Monte Carlo intervals for reversal frequencies; a deterministic influence curve has no such interval. Sampling uncertainty instead resamples comparison clusters and refits unedited data 500 times: Arena battle IDs, MT-Bench questions, and NBA dates. The initial original pair stays fixed for primary bootstrap margins. Percentile intervals are conditional descriptive summaries, without simultaneous or post-selection coverage guarantees. Arena has no available user identifiers; question clustering does not model arbitrary crossed annotator dependence; NBA dates do not remove season drift.

4 Data and experimental design

We use fixed public versions and retain all participants in each primary dataset. Arena contains 57,477 unique battles among 64 models, including 17,761 ties. Incident comparison counts range from 100 to 7,387; 1,275 of 2,016 possible unordered pairs are observed. MT-Bench contains 3,355 human judgments and 2,400 GPT-4 judgments on 80 questions, two turns, and six models. Human labels combine expert and author annotations with variable replication. The GPT-4 file covers every question–pair–turn cell, including 380 outcomes marked "tie (inconsistent)"; primary fitting treats these and ordinary ties as half-wins. Total ties are 780 human and 600 GPT-4 judgments.

For sports, we use the public FiveThirtyEight records [FiveThirtyEight, 2025], deduplicate the two recorded sides of each game using the supplied copy indicator, keep NBA regular seasons ending in 2006–2015, and fit each season separately. Each has 30 teams; eight seasons contain 1,230 games, 2012 has 990, and 2013 has 1,229. No ties occur. This is a static season-strength model fitted to results, not the source's published Elo ranking.

The primary analyses comprise 13 rankings with the same fitting/intervention conventions. We then vary λ to 0.1 and 10; restrict Arena entrants to at least 500 original incident comparisons; remove Arena ties; retain expert-only human judgments; or exclude inconsistent GPT-4 labels. A separate NBA control estimates a home-advantage coefficient and uses 300 date bootstraps. Adjacent-pair diagnostics extend the sensitivity search beyond first place.

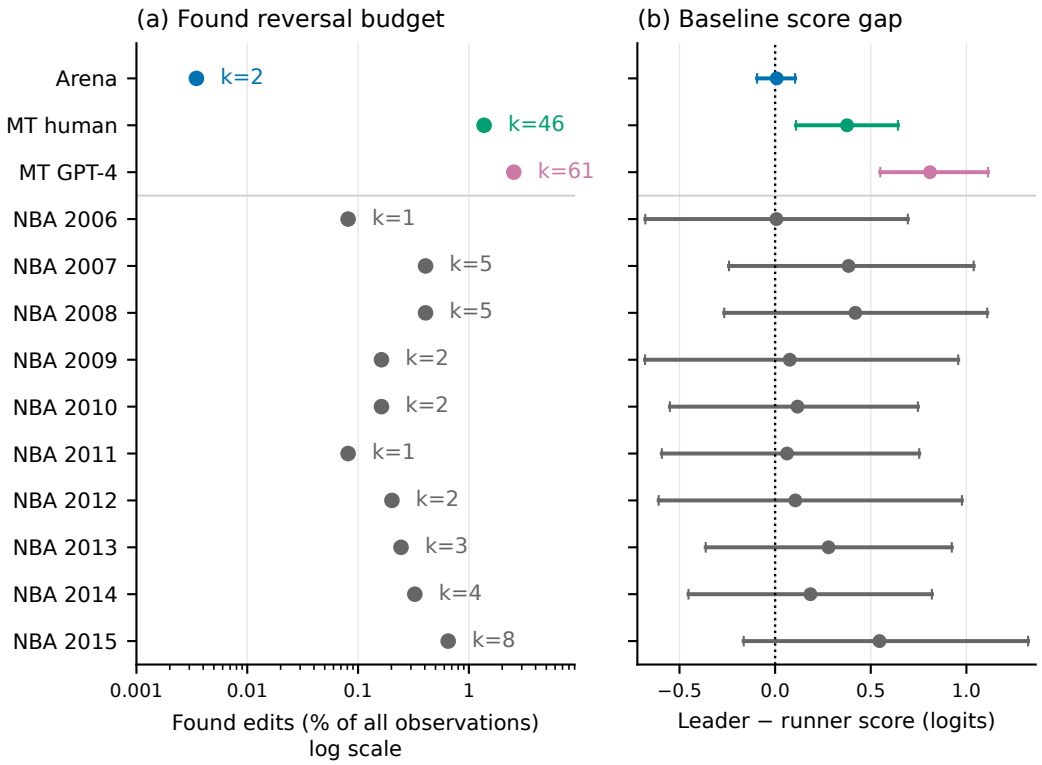


Figure 1: Small found attack budgets coincide with unresolved baseline leading contrasts in these fits. (a) Successful influence-prefix attacks on each original leading pair, shown as a percentage of all observations with absolute edit counts. These are upper bounds on the smallest successful attack, not certified minima. (b) Unperturbed score gaps and 95% percentile ranges from 500 cluster-bootstrap replicates, with the original pair held fixed. Clusters are Arena battles, MT questions, and NBA dates. Ranges are conditional descriptive uncertainty, without simultaneous or post-selection coverage guarantees. NBA seasons are separate fits; scores are in logit units.

For a closer human/model comparison, we intersect canonical question–pair–turn cells. On each of 1,814 shared cells, we select one observed human judgment uniformly and retain its label alongside GPT-4's label. We repeat human selection 100 times. A joint 500-replicate control resamples the same 80 questions for both label sources and samples one human annotation per cell before expanding the shared draw. Its first statistic reselects the top two in each fit. Because the human runner-up changes frequently, we add a post-hoc analysis fixing GPT-4 versus Claude-v1, the original leading pair in both full datasets. A resample that already reverses this reference pair has zero additional cost. These two statistics answer different questions and are reported separately.

5 Results

5.1 Found budgets and unedited uncertainty

Figure 1 places the successful budgets alongside unedited leading-pair uncertainty. The leading contrasts reverse after 2 Arena edits (0.00348%), 46 human MT-Bench edits (1.371%), and 61 GPT-4 MT-Bench edits (2.542%). NBA found counts, in chronological order, are $1, 5, 5, 2, 2, 1, 2, 3, 4, 8$, or 0.0813–0.6504% of the corresponding season. Every count has a saved row-level witness and an exact-refit verification. These results demonstrate the existence of small successful changes; they do not estimate globally minimum budgets.

The uncertainty context changes their interpretation. Arena's original logit gap is 0.00813, with a 95% bootstrap percentile interval $[-0.09410, 0.10430]$; its original leader loses first place in 45.4% of unedited bootstrap fits. Every NBA leading-pair interval also spans zero. The human and GPT-4

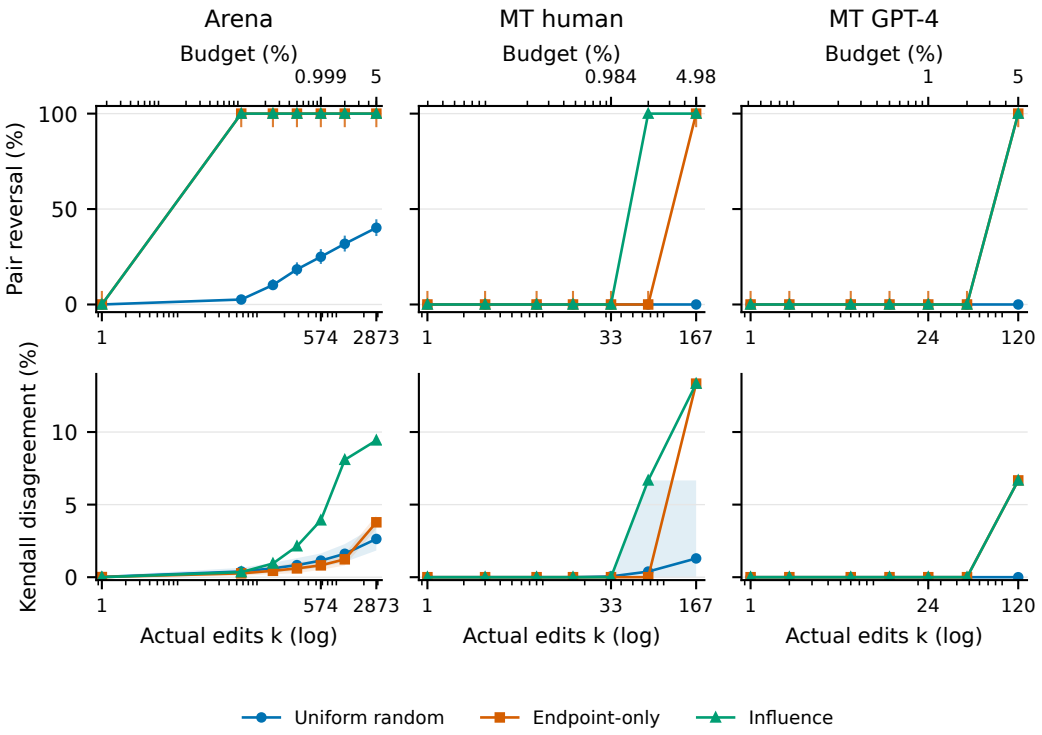


Figure 2: Targeting reverses leading contrasts more often than random edits while producing a different degree of whole-ranking change. The upper row shows fixed original leader–runner reversal rates; vertical bars are 95% exact binomial Monte Carlo intervals across 500 uniform-random or 50 endpoint-only runs. Deterministic influence attacks have no Monte Carlo interval. The lower row shows mean normalized Kendall pairwise disagreement; shading gives empirical 2.5–97.5 percentiles across perturbation runs, not data-bootstrap uncertainty. Markers are measured integer budgets; connecting lines guide the eye. Top axes convert counts to percentages of all observations. Every displayed strategy uses its full budget; exact budget mappings appear in the appendix. Reversing the original leading pair displaces the original winner, but the original runner-up need not become the new winner. A winner change can also occur without reversal of that pair.

MT-Bench intervals are $[0.10956, 0.64196]$ and $[0.54993, 1.11269]$, respectively. Thus the smallest found budgets accompany baseline leading contrasts that are already unresolved under the chosen resampling scheme. This is an observed association across these rankings, not a causal explanation or coverage guarantee.

5.2 Targeting, random noise, and whole-rank change

Figure 2 shows the measured budget curves; the full NBA panel is in Appendix E. At budgets near 1% of all rows, both targeted strategies reverse Arena and all ten NBA original leading pairs. Random reversals do so in 25.0% of Arena runs (Monte Carlo 95% interval 21.3–29.0%) and 0–38.2% of NBA runs. Neither MT-Bench leading pair reverses under any tested strategy at this budget. Zero of 500 random runs has an exact binomial upper endpoint of 0.735%; this concerns Monte Carlo frequency, not immunity to other edits.

First place and global order need different reporting. At the same Arena budget, influence targeting gives $D = 0.0392$, endpoint targeting gives mean $D = 0.00816$, and uniform random reversal gives mean $D = 0.0114$. A targeted strategy can therefore reverse first place while changing fewer other orderings than random noise. In NBA 2006, random leading-pair reversal occurs in 38.2% of runs while any winner change occurs in 38.6%; even these closely related endpoints are not identical. Budget fractions are rounded down to integer edits, so "near 1%" is nine games in the 990-game season, versus 12 in most other seasons.

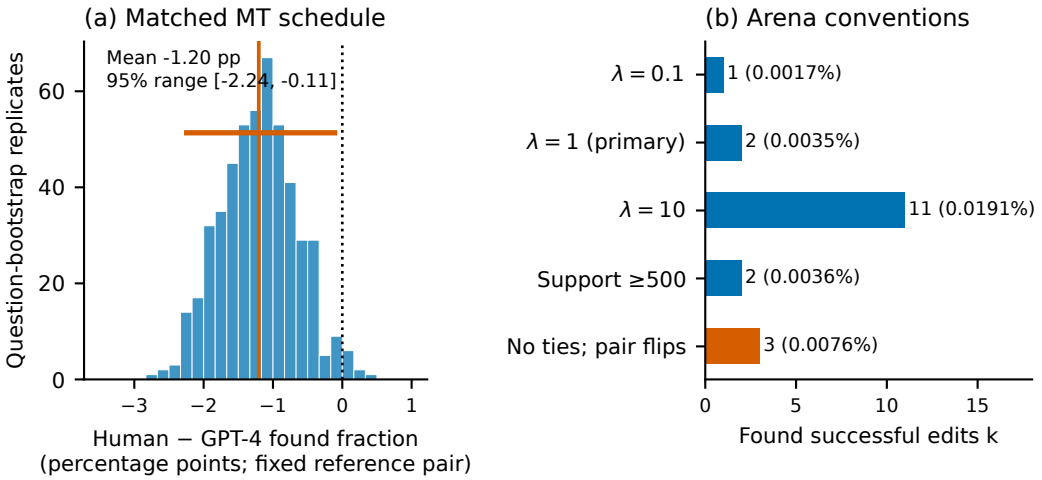


Figure 3: The matched MT contrast persists on shared support, while Arena counts depend on fitting conventions. (a) Human minus GPT-4 found attack fractions on the matched MT schedule, holding the GPT-4–Claude-v1 reference pair fixed. The histogram contains 500 paired question-bootstrap replicates with human-label resampling; orange marks show the mean and 95% percentile range (percentage points). This reference-pair analysis is a post-hoc follow-up; five human resamples already reverse the pair and require zero additional edits. (b) Arena found counts under changed conventions; labels give percentages using each fit's own observation count. Removing ties swaps the initial leader and runner, so its attack addresses the newly selected contrast. These controls describe the observed snapshots and heuristic, without identifying a causal evaluation-design effect or a globally minimal attack budget.

5.3 A shared-schedule human/model contrast

Figure 3 displays the fixed-reference matched contrast and Arena convention checks. On the fixed 1,814-cell support, every one of 100 human-label selections yields a smaller found top-two reversal fraction than GPT-4. Their mean difference is -1.152 percentage points. This variation conditions on the observed questions and concerns which human annotation is selected.

The joint question bootstrap gives a mean difference of -1.243 percentage points for the reselected-top-two statistic, with a percentile range $[-2.278, -0.110]$ points. However, GPT-3.5 replaces Claude-v1 as human runner-up in 106 of 500 replicates while GPT-4 remains leader; the human leader also changes in six. The post-hoc fixed GPT-4–Claude-v1 contrast avoids comparing a changing pair: its mean human-minus-GPT-4 found fraction is -1.205 percentage points, with range $[-2.244, -0.110]$. Humans have lower fractions in 98.2% of replicates; five human replicates already reverse the reference pair and receive cost zero. All searches complete without censoring.

This shared-support result removes unequal row replication and observed schedule as explanations of this particular contrast. It still combines differences in labels, ties, fitted gaps and annotation dependence. Its percentile range describes a heuristic found-budget statistic; it is neither an interval for the optimal budget nor a general causal effect of human versus model judging.

5.4 Modeling choices and sensitivity beyond first place

Arena's found budget changes from one to two to eleven edits as λ changes from 0.1 to 1 to 10. Restricting original incident support to at least 500 gives two edits. Removing ties changes the initial Arena leader and gives a three-edit reversal for the newly selected pair. These controls qualify any headline count: they change the specified estimator or population, and decisive-only fitting changes the target itself.

MT-Bench found counts remain unchanged across the tested ridge values. Expert-only humans yield 41 of 2,668 edits; excluding inconsistent GPT-4 ties yields 47 of 2,020. These are separate subsets, not repeated estimates of one unchanged quantity. The NBA home-advantage control retains every

original leading pair and every found count, with all adjusted bootstrap intervals still spanning zero. Home adjustment addresses one omission; it does not remove temporal changes in strength.

The phenomenon also extends to other adjacent ranks. Exact refits demonstrate a one-edit reversal for 22 of Arena's 63 adjacent original pairs and 15–22 of 29 per NBA season. No MT-Bench adjacent pair has a single-edit reversal in the heuristic search. The positive counts establish existence; the negative search result does not certify absence.

6 Discussion and limitations

A useful audit should accompany a successful-edit count with its starting margin and sampling uncertainty. When an unedited pair is poorly resolved, a tiny witness can reveal the fragility of its displayed order without showing that the underlying evaluation is invalid. When a contrast is better separated, the same fitted model may require substantially more edits. Our results support this joint reporting practice on the studied snapshots, while leaving general prediction of sensitivity open.

Comparing equal fractions alone is insufficient. One flipped game represents a larger share of a small season than one Arena battle, and a tied outcome enters the denominator without being editable. Participant count, comparison topology, ridge strength relative to sample size, and fitted gaps all matter. The matched MT-Bench analysis makes one comparison more controlled, but does not identify a design-level causal effect. Nor does our selected-pair heuristic establish worst-case stability of an entire ranking. Exhaustive optimization or certified lower bounds would answer a stronger question and are not supplied.

Our intervention assumes the ability to revise existing decisive labels with full data access. Real attackers may submit new votes under authentication, sampling and rate constraints; accidental errors may follow correlated mechanisms rather than uniform flips. We evaluate neither setting directly. We also preserve all ties, use a static fractional-tie Bradley–Terry model, and study a limited collection of historical snapshots. Bootstrap clusters approximate available dependence and cannot recover missing users, fully crossed annotators, or evolving team strength. Selection of the original top pair and the post-hoc fixed-reference control limit interval interpretation.

These diagnostics can improve evaluation transparency, while the same targeting ideas could guide selective corruption where historical outcomes are editable. We release code and aggregate evidence for public snapshots, without implementing platform submissions or redistributing raw conversations and full source datasets. Minimal processed outcome rows support witness verification. The practical implication is to expose uncertainty and specification dependence alongside a rank, and to retain reproducible witnesses for concrete sensitivity claims. Sensitive witnesses should be understood as diagnostics under a declared intervention. They do not by themselves establish compromised votes, an exploitable platform, or that a different judging design would fix the observed sensitivity.

7 Conclusion

Across public preference and sports data, successful label reversals range from a few outcomes to several percent, and the smallest found budgets coincide with unresolved bootstrap leading contrasts. Shared-schedule MT-Bench results retain a descriptive difference between human and GPT-4 labels, while robustness controls show that a single sensitivity number depends on modeling choices. Explicit endpoints, exact refits, saved witnesses, and separate accounts of perturbation and sampling uncertainty make these conclusions checkable without treating heuristic searches as certificates.

References

Tamara Broderick, Ryan Giordano, and Rachael Meager. An automatic finite-sample robustness metric: When can dropping a little data make a big difference? *arXiv preprint arXiv:2011.14999*, 2020. URL https://arxiv.org/abs/2011.14999v5. Consulted version 5, July 2023.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios N. Angelopoulos, Tianle Li, Dacheng Li, Banghua Zhu, Hao Zhang, Michael I. Jordan, Joseph E. Gonzalez, and Ion Stoica. Chatbot Arena: An open platform for evaluating LLMs by human preference. *arXiv preprint arXiv:2403.04132*, 2024. URL https://arxiv.org/abs/2403.04132v1. Consulted version 1.

FiveThirtyEight. NBA elo data, 2025. URL https://github.com/fivethirtyeight/data/tree/4c1ff5e3aef1816ae04af63218015066e186c147/nba-elo. Pinned public data snapshot, CC-BY-4.0.

Jenny Y. Huang, Yunyi Shen, Dennis Wei, and Tamara Broderick. Dropping just a handful of preferences can change top large language model rankings. *arXiv preprint arXiv:2508.11847*, 2025. URL https://arxiv.org/abs/2508.11847v1. Consulted version 1.

Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, 2017. URL https://arxiv.org/abs/1703.04730v3. Consulted arXiv:1703.04730v3.

Rui Min, Tianyu Pang, Chao Du, Qian Liu, Minhao Cheng, and Min Lin. Improving your model ranking on Chatbot Arena by vote rigging. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 2025. URL https://arxiv.org/abs/2501.17858v2. Consulted arXiv:2501.17858v2.

Shivalika Singh, Yiyang Nan, Alex Wang, Daniel D'souza, Sayash Kapoor, Ahmet Üstün, Sanmi Koyejo, Yuntian Deng, Shayne Longpre, Noah A. Smith, Beyza Ermis, Marzieh Fadaee, and Sara Hooker. The leaderboard illusion. *arXiv preprint arXiv:2504.20879*, 2025. URL https://arxiv.org/abs/2504.20879v2. Consulted version 2.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *arXiv preprint arXiv:2306.05685*, 2023. URL https://arxiv.org/abs/2306.05685v4. Consulted version 4; NeurIPS 2023 Datasets and Benchmarks.

A Reproduction and provenance

The accompanying repository contains the complete numerical analysis in `analysis/`, rendered presentation assets and their generator in `presentation/`, and the official conference style files in `venue/`. The root `README.md` gives the build entry point and links the analysis and presentation instructions. Raw data are downloaded to ignored `analysis/data/raw/`; raw conversations and full source datasets are not redistributed with this paper. Minimal processed outcome rows, including participant labels and available question/judge identifiers, are distributed as exact attack witnesses.

Arena is pinned to dataset commit `18c298340948c0e7f7727399fd459cca6ce0ca6f`; MT-Bench to `f7d2896d2cc5d80f8b55c2bbc722613555233c25`; and the NBA source to repository commit `4c1ff5e3aef1816ae04af63218015066e186c147`. Exact file URLs, licenses, byte counts and SHA-256 hashes are in `analysis/results/provenance.json`. The parsers preserve canonical participant order, invert labels when swapping participants, validate source label categories, and remove duplicate NBA game representations. All human MT annotations are included in the primary human fit; the expert-only subset is a separately named control.

The recorded primary environment is Python 3.11.2, NumPy 2.4.6, SciPy 1.17.1 and pandas 3.0.6; the dependency file also pins PyArrow 26.0.0. Runs use one BLAS/OpenMP thread on CPU, with base seed 20261102 and explicit dataset/replicate offsets in the runner. `run.json` records each workload's arguments and package versions; per-ranking audits record measured elapsed times and optimization diagnostics. The primary workload totals 99.4 seconds of measured wall time on the shared host. This is a measured workload summary, not a universal runtime guarantee; downloads and other controls are additional. The robustness, human-label selection and joint matched-bootstrap workload sums are 33.1, 5.3 and 34.0 seconds, respectively. Jobs ran under a two-CPU, 4 GB memory ceiling with no GPU; these are allocation limits, not peak-use measurements. Detailed commands and timing records, including preliminary and orchestration failures, are in the analysis README and execution log.

The source-to-claim chain is explicit. `scripts/fetch_data.py` obtains pinned data; `datasets.py` parses them; `bt.py` fits and evaluates the model; `run_study.py` creates primary, robustness and matched analyses; `matched_fixedpair.py` implements the post-hoc reference contrast; `nba_home.py` fits the home control; `diagnostics.py` produces adjacent-pair checks; and `summarize.py` aggregates the recorded refits. These script paths are relative to `analysis/`. The rendering manifest identifies the source CSVs and generator for each displayed figure and table. Reported numerical summaries can be inspected without downloading raw data; rerunning fits requires those inputs.

B Exact witnesses and approximation checks

For each primary found budget, `analysis/results/witnesses/` records distinct processed-row indices, participant names, original labels and reversed labels. `verify_witnesses.py` reconstructs the data and refits the full objective after applying those rows. The Arena witness reverses two original wins by `gpt-4-1106-preview` over `gpt-4-0125-preview`, identified by battle IDs 175888324 and 244123213. Its exact leading contrast changes from 0.008134 to -0.000871. The human and GPT-4 MT witnesses give post-edit margins -0.007132 and -0.006431, respectively. These identify concrete changes sufficient for the stated fitted endpoint; they do not show that all smaller subsets fail.

The one-edit approximation follows by differentiating the stationarity equation. Reversing y_i changes the gradient by $-(1-2y_i)x_i$ at the original fit; applying the inverse Hessian gives $\Delta\theta \approx H^{-1}(1-2y_i)x_i$, yielding Equation 3. The Hessian uses all observations, including ties. For each primary ranking, representative influence-ordered and random decisive rows are checked by actual single-row refits. The delivered per-row checks and full-budget curve outputs retain both the linear prediction and exact contrast, so the approximation can be assessed rather than assumed valid for a large prefix. Explicit numerical checks compare an independent SciPy optimizer, orientation invariance, infinitesimal-label influence finite differences, and duplication/penalty scaling at $\lambda \in \{0.1, 1, 10\}$. Parser assertions validate source IDs, allowed labels and canonical participant order.

At a found budget, the neighboring smaller prefix is also refitted and recorded. Binary search is a heuristic even within this candidate ordering because no monotonicity result is assumed. Its successful output remains a valid witness regardless of whether a different prefix or subset succeeds earlier.

Attack fractions in matched bootstraps likewise concern this algorithm and cap. The fixed-reference control assigns cost zero when the unedited resample already has the opposite reference ordering; treating those draws as failed searches would change the estimand.

C Analysis sequence and scope

The empirical plan was developed during the task rather than preregistered. A measured pilot sized the full CPU runs; pilot outputs are retained separately and do not substitute for the final primary results. Primary fixed-pair bootstrap margins and the matched control's reselected-pair statistic were initially distinct. After inspecting frequent human runner-up changes in the latter, we added the explicitly post-hoc fixed GPT-4–Claude-v1 analysis. The report preserves both quantities and does not reinterpret a reselected-pair range as uncertainty for a single fixed contrast. The supplied analysis reports all planned analyses executed and no searches censored in the final paired control.

The study does not include deletion or injected-vote experiments, global optimal-budget certificates, online defenses, a dynamic sports model, or comparisons of present-day production leaderboards. These are outside the defined intervention and claims. No missing experiment is represented by illustrative or synthetic measurements.

D Research assistance, data use, and independence

AI assistance was used for literature synthesis, implementation, presentation, manuscript drafting and checking. The experimental labels are the original public human, GPT-4 and sports observations; no model was queried to generate replacement evaluation labels for this study. Outcome reversals are deterministic transformations for sensitivity analysis.

We use public data subject to their recorded licenses (Apache-2.0 for the selected Arena snapshot and CC-BY-4.0 for MT-Bench and the NBA source), cite their sources, and exclude raw text and full source files from the artifact while retaining minimal processed witness rows for verification. The paper studies aggregate sensitivity, not individual voters or annotators. A historical sensitivity witness is not evidence of misconduct by any participant.

Literature consultation was limited to permitted supplied sources and direct methodological context, without seeking, identifying or following the withheld study or its discussions. The context supply records the sources and sections actually read, and the analysis independence note records its direct dataset-only access. No prohibited encounter is reported by those supplies. This restriction limits the breadth of the related-work survey; it does not license a first-discovery or comprehensive-novelty claim.

E Complete display tables and season curves

Tables 1–6 retain the exact counts, target identities, source conventions and denominator mappings behind the main figures. Figure 4 shows every NBA season. All displays are generated by `presentation/render.py`; the input and output manifest is in `presentation/README.md`.

Table 1: Dataset coverage and original selected pairs. Arena names abbreviate `gpt-4-1106-preview` and `gpt-4-0125-preview`; MT uses `gpt-4` and `claude-v1`. Minimum support counts incident comparisons per entrant. Bootstrap clusters are battles for Arena, questions for MT, and dates for NBA. NBA labels are season-ending years.

Ranking	Original leader > runner	N	Ties (%)	Min support	Clusters
Arena	GPT-4-1106 > GPT-4-0125	57,477	30.9	100	57,477
MT human	GPT-4 > Claude-v1	3,355	23.2	996	80
MT GPT-4	GPT-4 > Claude-v1	2,400	25.0	800	80
NBA 2006	SAS > DET	1,230	0.0	82	164
NBA 2007	DAL > PHO	1,230	0.0	82	163
NBA 2008	BOS > LAL	1,230	0.0	82	162
NBA 2009	CLE > LAL	1,230	0.0	82	163
NBA 2010	CLE > ORL	1,230	0.0	82	164
NBA 2011	SAS > CHI	1,230	0.0	82	164
NBA 2012	SAS > CHI	990	0.0	66	120
NBA 2013	MIA > OKC	1,229	0.0	81	163
NBA 2014	SAS > OKC	1,230	0.0	82	163
NBA 2015	GSW > ATL	1,230	0.0	82	162

Table 2: Primary fixed-snapshot evidence. N counts all comparisons, including ties; M counts entrants. Found k is a successful influence-prefix attack and only an upper bound. Each gap is the original leader minus runner in logit units; brackets give the conditional 95% cluster-bootstrap percentile range (500 replicates).

Ranking	N	M	Decisive	Found k	$100k/N$ (%)	Gap [95% range]
Arena	57,477	64	39,716	2	0.0035	0.008 [-0.094, 0.104]
MT human	3,355	6	2,575	46	1.3711	0.376 [0.110, 0.642]
MT GPT-4	2,400	6	1,800	61	2.5417	0.809 [0.550, 1.113]
NBA 2006	1,230	30	1,230	1	0.0813	0.007 [-0.679, 0.694]
NBA 2007	1,230	30	1,230	5	0.4065	0.384 [-0.241, 1.038]
NBA 2008	1,230	30	1,230	5	0.4065	0.419 [-0.266, 1.109]
NBA 2009	1,230	30	1,230	2	0.1626	0.076 [-0.680, 0.957]
NBA 2010	1,230	30	1,230	2	0.1626	0.117 [-0.550, 0.746]
NBA 2011	1,230	30	1,230	1	0.0813	0.062 [-0.592, 0.753]
NBA 2012	990	30	990	2	0.2020	0.106 [-0.609, 0.976]
NBA 2013	1,229	30	1,229	3	0.2441	0.279 [-0.363, 0.924]
NBA 2014	1,230	30	1,230	4	0.3252	0.185 [-0.453, 0.820]
NBA 2015	1,230	30	1,230	8	0.6504	0.546 [-0.164, 1.322]

Table 3: Effects at $k = \lfloor 0.01N \rfloor$ actual edits. Random, Endpoint and Influence are fixed-pair reversal percentages; random and endpoint rates average 500 and 50 runs respectively. Winner is any change of global winner under the deterministic influence attack. Kendall is its normalized pairwise disagreement, expressed as a percentage. Rounding makes actual fractions slightly below 1% in some datasets; exact mappings are supplied separately.

Ranking	k	Random (%)	Endpoint (%)	Influence (%)	Winner (%)	Kendall (%)
Arena	574	25.0	100.0	100.0	100.0	3.92
MT human	33	0.0	0.0	0.0	0.0	0.00
MT GPT-4	24	0.0	0.0	0.0	0.0	0.00
NBA 2006	12	38.2	100.0	100.0	100.0	2.99
NBA 2007	12	0.0	100.0	100.0	100.0	2.53
NBA 2008	12	0.0	100.0	100.0	100.0	3.22
NBA 2009	12	13.0	100.0	100.0	100.0	2.30
NBA 2010	12	4.0	100.0	100.0	100.0	4.60
NBA 2011	12	22.8	100.0	100.0	100.0	2.30
NBA 2012	9	7.2	100.0	100.0	100.0	3.68
NBA 2013	12	0.2	100.0	100.0	100.0	3.22
NBA 2014	12	0.4	100.0	100.0	100.0	2.53
NBA 2015	12	0.0	100.0	100.0	100.0	2.53

Table 4: Convention sensitivity of found influence-prefix budgets. Ties enter the primary fit as half-wins. Regularization uses the summed likelihood plus $\lambda\|\theta\|^2/2$. Arena no-tie filtering swaps the initial leading pair; the attack is redefined on that fit's selected pair. Arena support filtering and MT filters also change N, which remains the denominator. Counts do not certify minimum edits or robustness below the found count.

Convention	N	M	Found k	$100k/N$ (%)
Arena: $\lambda = 0.1$	57,477	64	1	0.0017
Arena: $\lambda = 1$	57,477	64	2	0.0035
Arena: $\lambda = 10$	57,477	64	11	0.0191
Arena: support ≥ 500	54,985	55	2	0.0036
Arena: no ties (pair flips)	39,716	64	3	0.0076
MT human: $\lambda = 0.1$	3,355	6	46	1.3711
MT human: $\lambda = 1$	3,355	6	46	1.3711
MT human: $\lambda = 10$	3,355	6	46	1.3711
MT human: expert only	2,668	6	41	1.5367
MT GPT-4: $\lambda = 0.1$	2,400	6	61	2.5417
MT GPT-4: $\lambda = 1$	2,400	6	61	2.5417
MT GPT-4: $\lambda = 10$	2,400	6	61	2.5417
MT GPT-4: consistent only	2,020	6	47	2.3267

Table 5: Paired human minus GPT-4 found attack fractions on common MT support (percentage points). The first row varies human labels over 100 selections on 1,814 fixed cells; its range captures label selection, not question uncertainty. The next two rows use 500 joint question-bootstrap draws with matched counts and human-label resampling. Reselected-pair statistics choose each resample's top two; the post-hoc fixed-reference statistic uses GPT-4–Claude-v1 and assigns zero edits to already reversed contrasts. Ranges are empirical percentiles of heuristic statistics, not coverage guarantees for minimum budgets.

Matched statistic / resampling	Runs	Mean difference (pp)	95% range (pp)
Reselected pair / human labels	100	-1.15	[-1.38, -0.88]
Reselected pair / joint questions	500	-1.24	[-2.28, -0.11]
Fixed reference pair / joint questions	500	-1.20	[-2.24, -0.11]

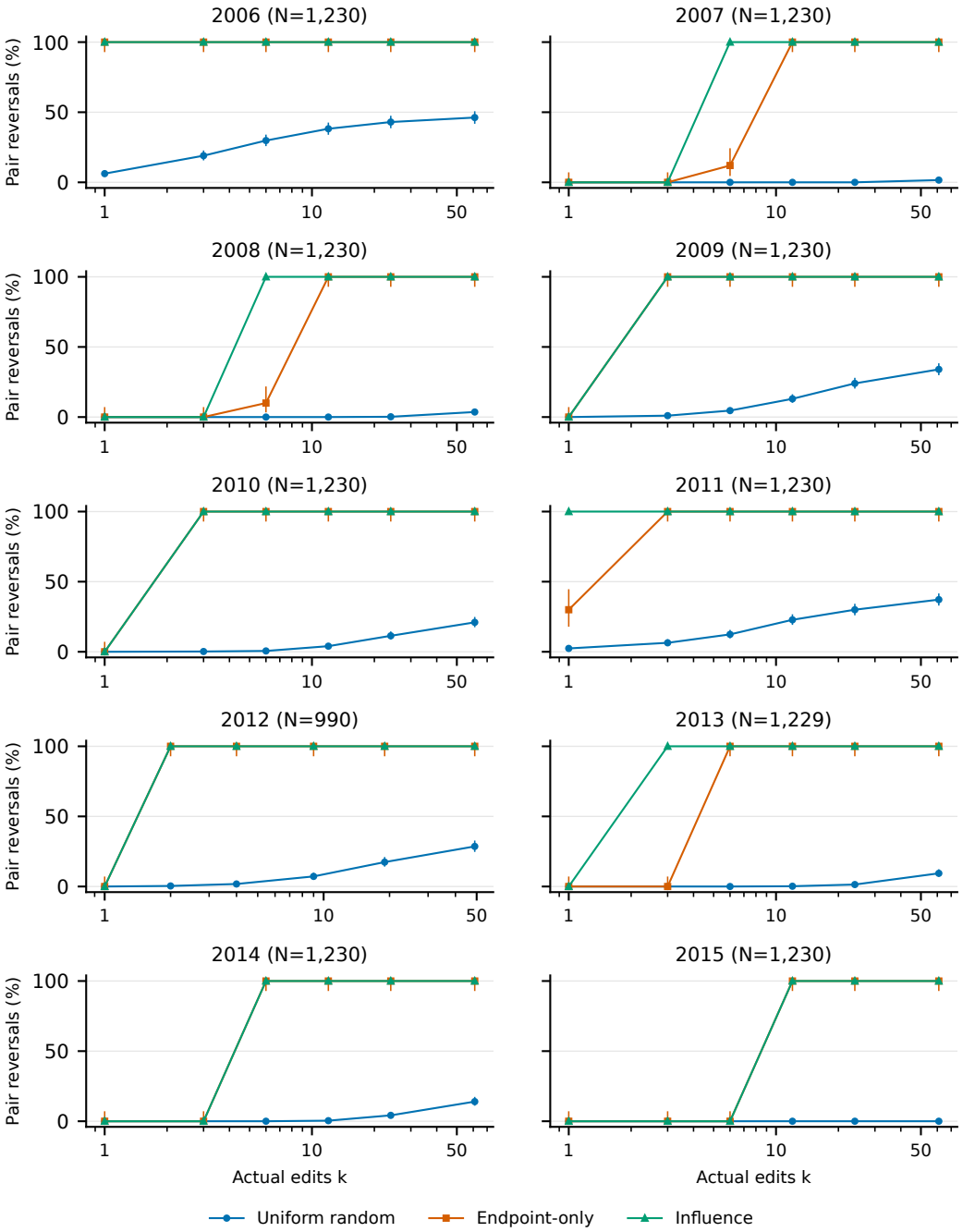


Figure 4: All ten NBA seasons under the same exact-refit perturbation protocol. Markers show actual integer edit counts on logarithmic axes; lines guide the eye between measured budgets. Vertical bars show 95% exact binomial Monte Carlo intervals for uniform random (500 runs) and endpoint-only (50 runs); the influence sequence is deterministic. Seasons have distinct original leading pairs and sample sizes. Reversal of the original leading pair implies that the original leader loses first place, but the new winner can be a third team. Winner change can also occur without reversal of that pair. Fractional budget mappings are reported separately.

Table 6: Exact integer budgets and their percentages of all observations. Nominal budgets are one edit and $\max(1, \lfloor fN \rfloor)$ for $f \in \{0.001, 0.0025, 0.005, 0.01, 0.02, 0.05\}$. Duplicate budgets are removed. For every primary dataset, strategy and replicate, actual edits equal the integer budget: no displayed primary curve saturates.

Ranking	Actual edits k (percentage of all observations)
Arena	1 (0.0017%); 57 (0.0992%); 143 (0.2488%); 287 (0.4993%); 574 (0.9987%); 1149 (1.9991%); 2873 (4.9985%)
MT human	1 (0.0298%); 3 (0.0894%); 8 (0.2385%); 16 (0.4769%); 33 (0.9836%); 67 (1.9970%); 167 (4.9776%)
MT GPT-4	1 (0.0417%); 2 (0.0833%); 6 (0.2500%); 12 (0.5000%); 24 (1.0000%); 48 (2.0000%); 120 (5.0000%)
NBA 2006	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2007	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2008	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2009	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2010	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2011	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2012	1 (0.1010%); 2 (0.2020%); 4 (0.4040%); 9 (0.9091%); 19 (1.9192%); 49 (4.9495%)
NBA 2013	1 (0.0814%); 3 (0.2441%); 6 (0.4882%); 12 (0.9764%); 24 (1.9528%); 61 (4.9634%)
NBA 2014	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)
NBA 2015	1 (0.0813%); 3 (0.2439%); 6 (0.4878%); 12 (0.9756%); 24 (1.9512%); 61 (4.9593%)

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